

✓ Key Components

- 1.StateGraph : The core of our application, defining the flow of our Travel Planner
- 2.PlannerState: A custom type representing the state of our planing process
- 3.Node Functions: Individual steps in our planing process (input_city, input_interests, create_itinerary)
- 4.LLM Integration: Utilizing a language model to generate the final itinerary.

```
!pip install langchain langchain_core langchain_groq langchain_community langchain langgraph
Collecting requests<2.34.2-py3-none-any.whl.metadata (4.8 kB)
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: sqlalchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain_community)
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Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->groq<1.0.0,>=0.30
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->groq
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Downloading pydantic_settings-2.14.2-py3-none-any.whl (61 kB)
  61.7/61.7 kB 5.6 MB/s eta 0:00:00
Downloading requests-2.34.2-py3-none-any.whl (73 kB)
  73.1/73.1 kB 6.2 MB/s eta 0:00:00
Downloading langchain_text_splitters-1.1.2-py3-none-any.whl (35 kB)
Installing collected packages: requests, httpx-sse, pydantic-settings, groq, langchain-text-splitters, langchain_groq, langcha
  Attempting uninstall: requests
    Found existing installation: requests 2.32.4
    Uninstalling requests-2.32.4:
      Successfully uninstalled requests-2.32.4
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is tl
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.34.2 which is incompatible.
Successfully installed groq-0.37.1 httpx-sse-0.4.3 langchain-classic-1.0.8 langchain-text-splitters-1.1.2 langchain_community-0.4.2
```

```
import os
from typing import TypedDict, Annotated, List
from langgraph.graph import StateGraph, END
from langchain_core.messages import HumanMessage, AIMessage
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.runnables.graph import MermaidDrawMethod
from IPython.display import display, Image
```

Define Agent

```
class PlannerState(TypedDict):
    messages : Annotated[List[HumanMessage | AIMessage], "the messages in the conversation"]
    city: str
    interests: List[str]
    itinerary: str
```

```
from langchain_groq import ChatGroq
llm = ChatGroq(
    temperature = 0,
    groq_api_key = "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",
    model_name = "llama-3.3-70b-versatile"
)
```

'A Multi-Agent System (MAS) is a system composed of multiple autonomous agents that interact with each other and their environment to achieve common or individual goals. In the context of Artificial Intelligence (AI), a Multi-AI Agent refers to a system that consists of multiple AI agents that work together to solve complex problems or perform tasks.\n\nIn a Multi-AI Agent system, each agent is a separate entity with its own capabilities, goals, and decision-making processes. These agents can be homogeneous (similar) or heterogeneous (different) in terms of their architecture, functionality, and objectives. The agents interact with each other through communication protocols, such as message passing or shared knowledge bases, to coordinate their actions

```
itinerary_prompt = ChatPromptTemplate.from_messages([
    ("system", "You are a helpful travel assistant. Create a day trip itinerary for {city} based on the user's interests: {interests}"),
    ("human", "Create an itinerary for my day trip."),
])
```

Define agent Function

```
def input_city(state: PlannerState) -> PlannerState:
    print("Please enter the city you want to visit for your day trip: ")
    user_message = input("Your Input: ")
    return {
        **state,
        "city": user_message,
        "messages": state['messages'] + [HumanMessage(content=user_message)]
    }

def input_interest(state: PlannerState) -> PlannerState:
    print(f"Please enter your interest for the trip to : {state['city']} (comma-separated): ")
    user_message = input("Your Input: ")
    return {
        **state,
        "interests": [interest.strip() for interest in user_message.split(",")],
        "messages": state['messages'] + [HumanMessage(content=user_message)]
    }

def create_itinerary(state: PlannerState) -> PlannerState:
    print(f"Creating an itinerary for {state['city']} based on interests : {' , '.join(state['interests'])}")
    response = llm.invoke(itinerary_prompt.format_messages(city = state['city'], interests = ' , '.join(state['interests'])))
    print("\nFinal Itinerary: ")
    print(response.content)
    return {
        **state,
        "messages": state['messages'] + [AIMessage(content=response.content)],
        "itinerary": response.content,
    }
```

Create and Compile the Graph

```
workflow = StateGraph(PlannerState)

workflow.add_node("input_city", input_city)
workflow.add_node("input_interest", input_interest)
workflow.add_node("create_itinerary", create_itinerary)

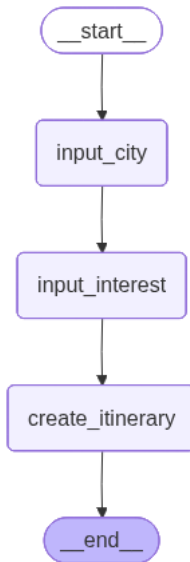
workflow.set_entry_point("input_city")

workflow.add_edge("input_city", "input_interest")
workflow.add_edge("input_interest", "create_itinerary")
workflow.add_edge("create_itinerary", END)
```

```
app = workflow.compile()
```

Display the graph structure

```
display(
    Image(
        app.get_graph().draw_mermaid_png(
            draw_method = MermaidDrawMethod.API
        )
    )
)
```



Define the function that runs the graph

```
def travel_planner(user_request: str):
    print(f"Initial Request: {user_request}\n")
    state = {
        "messages": [HumanMessage(content=user_request)],
        "city": "",
        "interests": [],
        "itinerary": "",
    }
    for output in app.stream(state):
        pass
```

```
user_request = "I want to plan a day trip"
travel_planner(user_request)
```

Initial Request: I want to plan a day trip

Please enter the city you want to visit for your day trip:

Your Input: Kandy

Please enter your interest for the trip to : Kandy (comma-separated):

Your Input: Temple of the Sacred Tooth Relic , Bahiravokanda Vihara Buddha Statue,

Creating an itinerary for Kandy based on interests : Temple of the Sacred Tooth Relic, Bahiravokanda Vihara Buddha Statue,

Final Itinerary:

Here's a brief, bulleted itinerary for your day trip to Kandy:

- * 8:00 AM: Start the day with a visit to the **Temple of the Sacred Tooth Relic**, a UNESCO World Heritage Site and one of the most sacred Buddhist sites in the world.
- * 10:00 AM: Take a short walk to the **Kandy Lake**, a scenic spot for relaxation and photography.
- * 11:30 AM: Head to the **Bahiravokanda Vihara Buddha Statue**, a 88-foot tall Buddha statue that offers stunning views of the city.
- * 1:00 PM: Take a break for lunch at a local restaurant, trying some traditional Sri Lankan cuisine.
- * 2:30 PM: Explore the **Kandy City**, visiting the Kandy Museum, the Royal Botanical Gardens, or taking a stroll through the city streets.
- * 5:00 PM: End the day with a **Cultural Dance Show**, showcasing traditional Sri Lankan music and dance.

This itinerary provides a mix of culture, history, and natural beauty, and can be adjusted based on your interests and preferences.

```
!pip install gradio
```

Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (6.19.0)

Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.14.0)

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Requirement already satisfied: fastapi<1.0,>=0.115.2 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.138.0)

Requirement already satisfied: gradio-client==2.5.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.5.0)

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Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)

Requirement already satisfied: huggingface-hub<2.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.20.1)

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Requirement already satisfied: numpy<3.0,>=1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.0.2)

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Requirement already satisfied: pillow<13.0,>=8.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)

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Requirement already satisfied: pydub<1.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.25.1)

Requirement already satisfied: python-multipart<1.0,>=0.0.18 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.0.32)

Requirement already satisfied: pytz>=2017.2 in /usr/local/lib/python3.12/dist-packages (from gradio) (2025.2)

Requirement already satisfied: pyyaml<7.0,>=5.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (6.0.3)

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Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)

Requirement already satisfied: starlette<2.0,>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (1.3.1)

Requirement already satisfied: tomkit<0.15.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.14.0)

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Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.18)

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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2026.6.17)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.14.0)

Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.2.0->gradio) (8.1.8)

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Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.2.0->gradio) (1.2.0)

Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.2.0->gradio) (4.67.1)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<4.0,>=1.0->gradio) (2.9.0.post0)

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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<4.0,>=1.0->gradio) (1.17.0)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<1.0,>=0.12->gradio) (3.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<1.0,>=0.12->gradio) (2.19.0)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer<1.0,>=0.12->gradio) (0.1.2)

```
import gradio as gr
from typing import TypedDict, Annotated, List
from langgraph.graph import StateGraph, END
from langchain_core.messages import HumanMessage, AIMessage
from langchain_core.prompts import ChatPromptTemplate
from langchain_groq import ChatGroq

class PlannerState(TypedDict):
    messages: Annotated[List[HumanMessage | AIMessage], "The messages in the conversation"]
    city: str
    interests: List[str]
    itinerary: str

# Define the LLM
llm = ChatGroq(
    temperature=0,
    groq_api_key="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",
    model_name="llama-3.3-70b-versatile"
)

# Define the itinerary prompt
itinerary_prompt = ChatPromptTemplate.from_messages([
    ("system", "You are a helpful travel assistant. Create a day trip itinerary for {city} based on the user's interests: {interests}."),
    ("human", "Create an itinerary for my day trip."),
])

def input_city(city: str, state: PlannerState) -> PlannerState:
    return {
        **state,
        "city": city,
    }
```

```

        "messages": state['messages'] + [HumanMessage(content=city)],
    }

def input_interests(interests: str, state: PlannerState) -> PlannerState:
    return {
        **state,
        "interests": [interest.strip() for interest in interests.split(',')],
        "messages": state['messages'] + [HumanMessage(content=interests)],
    }

def create_itinerary(state: PlannerState) -> str:
    response = llm.invoke(itinerary_prompt.format_messages(city=state['city'], interests=", ".join(state['interests'])))
    state["itinerary"] = response.content
    state["messages"] += [AIMessage(content=response.content)]
    return response.content

# Define the Gradio application
def travel_planner(city: str, interests: str):
    # Initialize state
    state = {
        "messages": [],
        "city": "",
        "interests": [],
        "itinerary": "",
    }

    # Process the city and interests inputs
    state = input_city(city, state)
    state = input_interests(interests, state)

    # Generate the itinerary
    itinerary = create_itinerary(state)

    return itinerary

# Build the Gradio interface
interface = gr.Interface(
    fn=travel_planner,
    theme='Yntec/HaleyCH_Theme_Orange_Green',
    inputs=[
        gr.Textbox(label="Enter the city for your day trip"),
        gr.Textbox(label="Enter your interests (comma-separated)"),
    ],
    outputs=gr.Textbox(label="Generated Itinerary"),
    title="Travel Itinerary Planner",
    description="Enter a city and your interests to generate a personalized day trip itinerary."
)

# Launch the Gradio application
interface.launch()

```

```
/usr/local/lib/python3.12/dist-packages/gradio/interface.py:171: UserWarning: The parameters have been moved from the Blocks constructor to the GradioInterface constructor. Please update your code to use the new signature.
  super().__init__(
```

theme_schema%400.0.1.json: 100%

12.6k/12.6k [00:00<00:00, 852kB/s]

It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share=True`.

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://32e427d72b16657b89.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `gradio --server-name`

Travel Itinerary Planner

Enter a city and your interests to generate a personalized day trip itinerary.

Enter the city for your day trip

Kandy

Enter your interests (comma-separated)

Temple of the Sacred Tooth Relic , Udawattekele Sanctuary, Kandy National Museum

Clear

Submit

Generated Itinerary

Here's a brief, bulleted itinerary for your day trip to Kandy:

- * 8:00 AM - 9:30 AM: Visit the Temple of the Sacred Tooth Relic, a UNESCO World Heritage Site and one of Buddhism's most sacred sites
- * 10:00 AM - 12:00 PM: Explore the Udawattekele Sanctuary, a forest reserve with historic and cultural significance, and enjoy a leisurely walk amidst nature
- * 1:00 PM - 3:00 PM: Have lunch and visit the Kandy National Museum to learn about the history and culture of Kandy and Sri Lanka
- * 3:30 PM - 5:00 PM: Take a stroll around the Kandy Lake and enjoy the city's scenic views before departing

